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ORIGINAL ARTICLE

The risk of chronic kidney disease or proteinuria with long or short sleep duration: a systematic review and meta-analysis of cohort studies

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Abstract

Objective Irregular sleep duration has been linked with systemic diseases as well as chronic kidney disease (CKD). However, most of the evidence is low-quality and from cross-sectional data. We hereby present a meta-analysis of cohort studies examining the longitudinal association between short and long sleep with the risk of CKD or proteinuria.

Methods Databases of Embase, PubMed, CENTRAL, Web of Science, and Scopus were searched up to 5th April 2024. The risk of CKD/proteinuria was assessed with short or long sleep duration.

Results Nine studies were included. Both short and long sleep duration were associated with a mild increase in the risk of CKD/proteinuria. Based on different cutoffs for short sleep, we noted that sleep of ≤ 7 h was not associated with a significantly increased risk of CKD/proteinuria. A mild significant risk was noted in the subgroup of ≤ 6 h while a significant association was noted for sleep ≤ 5 h. For longer sleep duration, individuals with ≥ 8 h of sleep had an increased risk of CKD/proteinuria. However, the results were non-significant for individuals with ≥ 9 h of sleep. Non-significant results were noted for separate analyses on male, female, high body mass index, and elderly (≥ 60 years) individuals.

Conclusion Both short and long sleep durations are associated with a significant increase in the risk of CKD/proteinuria in the adult population.

Keywords Sleep · Insomnia · Renal failure · Kidney disease · Pooled analysis

Introduction

Sleep is a crucial natural process that is essential in maintaining systemic health. Ample evidence suggests that sleep duration is the most important sleep component contributing to general well-being [1]. Based on the American Academy of Sleep Medicine, the Sleep Research Society, and the US National Sleep Foundation recommendations, sleep of 7–9 h (h) per day is judged to be optimal, whereas any sleep of ≤ 6 h per day is considered suboptimal [2, 3]. In recent times, there has been an increasing trend of short sleep duration, especially among working adults with data suggesting that about 1/3rd of the population is sleep-deprived [4, 5]. A large body of evidence shows that short sleep duration is associated with numerous lifestyle-related diseases such as obesity, diabetes, hypertension, cardiovascular diseases, and coronary artery disease, all of which can contribute to a higher risk of mortality [1, 6–9]. Similarly, meta-analytic studies have demonstrated that longer sleep duration is also linked to increased mortality and higher risk of cardiovascular events, diabetes, and obesity [9, 10].

Like poor sleep, chronic kidney disease (CKD) has emerged as a major public health issue worldwide. Recent data show that about 9.5% of the global population suffers from CKD [11]. Trends over the past 3 decades show that the prevalence of CKD and CKD-related deaths has gradually increased and it is now the 10th most common cause of death [12]. CKD not only increases the risk of end-stage kidney disease (ESKD) which requires dialysis and kidney

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replacement therapy, but also predicts cardiovascular disease and mortality [13, 14]. Despite the high morbidity and mortality of CKD, early detection and modification of relevant risk factors can slow or prevent the disease especially in high-risk groups [15]. One such modifiable risk factor could be sleep. Several cross-sectional studies have shown that poor sleep duration and quality is a risk factor for CKD [16–18]. Inadequate sleep can increase systemic inflammation which is a recognized risk factor for CKD [19, 20]. In the past, systematic reviews have shown that both short and long sleep duration can increase the risk of CKD and proteinuria in adults [21, 22]. However, most of the studies included in these reviews were cross-sectional studies which produce low-quality evidence. The causal relationship between exposure (sleep duration) and outcome (CKD) can only be reliably obtained by good-quality cohort studies with long follow-ups [23]. Therefore, to provide more reliable evidence, the present systematic review and meta-analysis were designed to assess the relationship between short and long sleep duration with CKD and proteinuria by collating data from cohort studies.

Materials and methods

Search methodology

This work is performed following PRISMA [24] and MOOSE guidelines [25]. The study was registered on PROSPERO under the number CRD42024528965. A detailed and comprehensive systematic search was performed by two reviewers with the help of an experienced medical librarian on Embase, PubMed, CENTRAL, Web of Science, and Scopus. All peer-reviewed articles irrespective of the date of publication and language were searched provided they were online till 5th April 2024. We also conduct a manual search on Google Scholar to ensure comprehensive coverage of gray literature. All databases were explored using the search query "(((sleep duration) OR (sleep time)) OR (insomnia)) AND (((((proteinuria) OR (chronic kidney disease)) OR (renal disease)) OR (end-stage renal disease)) OR (dialysis))".

The search outcomes of all repositories were collated and then deduplicated in reference manager software (EndNote, Clarivate, USA). Primary eligibility was assessed by reading the titles and abstracts of the unique articles. Two investigators were involved in the study screening. Irrelevant articles were eliminated and potentially eligible studies were acquired for further review. The investigators read the full texts and selected eligible articles for this review. We also inspected the references of all studies to identify further studies. Any non-English language studies were translated using Google Translate.

Eligibility criteria

The current review included studies which were (a) cohort in design and conducted on adult human population (b) examined the association between sleep duration (long or short) and subsequent risk of CKD or proteinuria (c) reported the association as a multivariate-adjusted ratio with 95% confidence intervals (CI). We did not predefine optimal, short, or long sleep duration and left it to the studies to establish cutoffs. We also did not define the cutoff of the estimated glomerular filtration rate (eGFR) for CKD or proteinuria for the inclusion criteria. Studies were included if they used any eGFR cutoff < 60 ml/min per 1.73 m^2 or measured a decrease in eGFR or used any method to determine proteinuria as a marker for CKD.

Case–control studies, cross-sectional studies, studies not reporting on sleep duration, reporting only univariate data, reporting only Kaplan–Meier curves and not numerical data, and published only as abstracts were excluded. If two articles reported duplicate sample sizes, the study with the larger sample size was included. If two articles had the same sample size, the study reporting the better definition of CKD was to be included.

Quality of studies

The Newcastle Ottawa Scale was used to examine the risk of bias in the cohort studies [26]. The NOS evaluates three study parameters and the possible total points are 4 points for selection, 2 points for comparability, and 3 points for outcomes. The quality of the study is divided into 3 groups with ≥ 8 points being considered as "good", 6 to 7 points as "fair", and ≤ 6 points as "poor" quality. Quality assessment was conducted independently by two authors, and a consensus of any disagreement was discussed with the senior author for a solution.

Data management

Relevant data extracted from studies included: the name of the author, database used, location, study period, sample size, demographic details, classification of sleep duration, CKD definition, assessment of sleep duration and CKD/proteinuria, individuals with short and long sleep duration, incidence of CKD, follow-up, adjusted factors, and details of subgroup analysis. All data were sourced by two authors in a pre-formatted table and cross-checked for any mistakes. In case any outcome data were missing, the corresponding author was to be contacted once via email.

We primarily examined the risk of CKD or proteinuria with short or long sleep duration vs optimal sleep duration

as defined by the studies. Secondarily, we also aimed to assess the risk of CKD/proteinuria with specific cutoffs of sleep duration, namely ≤ 5 h (h), ≤ 6 h, and ≤ 7 h for a short sleep and ≥ 8 h and ≥ 9 h for long sleep. Any other subgroup analysis based on demographic data was also planned depending on the reporting of data by the included studies. For subgroup analysis based on age, elderly was defined as ≥ 60 years.

Statistical analysis

Meta-analysis was conducted using Review Manager 5.3 (Cochrane Collaboration, 2014). Due to the low incidence of CKD in the cohorts, risk ratios (RR), incident rate ratios, and hazard ratios were crudely regarded as odds ratios (OR) [22]. All study-specific estimates were combined using inverse variance-weighted averages of logarithmic ORs in a random-effects model. If the study reported different ratios for subgroups, we combined the different groups into a single group of short or long sleep duration using the review manager software.

Heterogeneity among studies was quantified by obtaining the I value and the Chi-square test. $P < 0.05$ was considered statistically significant and an I^2 value of 50% or more indicated the presence of heterogeneity. Sensitivity analysis was conducted for the primary analysis to assess the contribution of each study to the pooled estimate by excluding individual studies one at a time and recalculating the pooled HR estimates for the remaining studies. The reanalysis of data was done in the software itself. Publication bias was assessed by the funnel plot and Egger's test. Subgroup analysis was conducted based on study design, location, CKD definition, and CKD assessment.

Results

Search results

The reviewers obtained 8028 articles from all databases. Of these, 3522 were overlapping and hence removed. 4506 studies were screened by reading the title and abstracts. 4451 articles were not relevant to the review and hence excluded. The remaining 55 were selected for further analysis. Forty-six studies were excluded with reasons. Nine studies [27–35] finally met the inclusion criteria (Fig. 1). No disagreements were recorded between reviewers for inclusion of the studies. No additional studies were identified on gray literature and reference search.

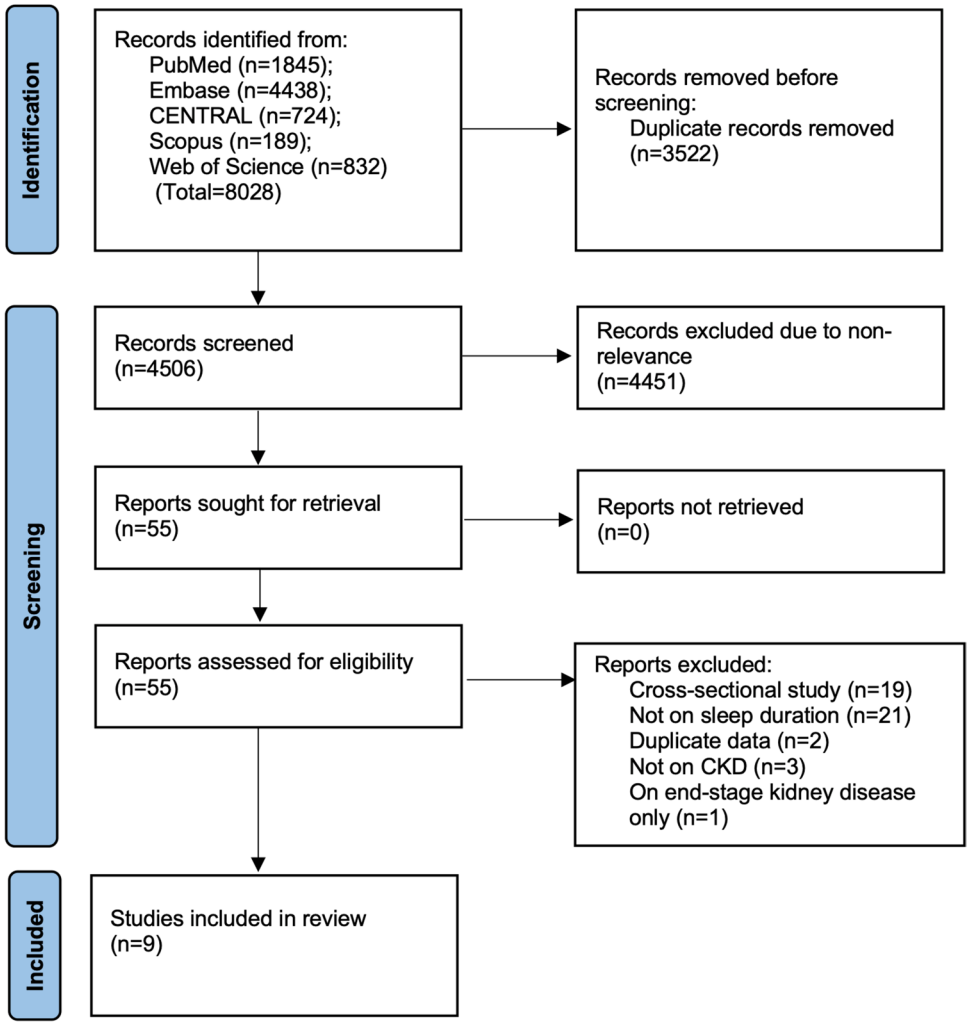
Baseline information

Study details extracted by the authors are presented in Table 1. The majority of studies were from Asia originating from China, Japan, Taiwan, Singapore, and Korea. Only two studies reported Western data, one from the USA and one from the UK. The study period was wide and included participants from 1976 to 2018. Two studies were retrospective while all others were prospective. All studies excluded patients with baseline CKD. The cumulative sample size of all studies was 671,517. All studies used self-reported questionnaires to assess sleep duration. Optimal sleep duration varied among studies and was defined as either 7–8 h, 7–9 h, 7 h, 6–9 h, 6–8 h, or 6–7 h. Accordingly, the definition of sort and long sleep also varied in the included studies. The incidence of CKD/proteinuria was 20,689 amounting to 3.08% of the total sample. The majority of studies used clinical measurements to assess CKD/proteinuria. Baseline eGFR was reported by only five studies. Geng et al. [33], Bo et al. [30], McMullan et al. [29], Sasaki et al. [28], and Yamamoto et al. [27] reported a mean baseline eGFR of > 90, 86.3, 88.3, > 84, and 90 ml/min per 1.73 m^2 for their cohorts, respectively. Two studies used medical records while one used a questionnaire. One study assessed only proteinuria while one included CKD and proteinuria as the outcome. One study did not report the exact follow-up duration. In the remaining studies, the follow-up ranged from 4.4 years to > 16 years. Assessing the study quality, the reviewers found the majority of studies to be of high quality achieving an NOS score of 8. Two studies received a score of 7. All studies assessed the risk of CKD/proteinuria with both short and long sleep duration, except for Yamamoto et al. [27] who reported only short sleep duration. Outcome ratios and any subgroup analysis conducted by the studies are presented in Supplementary Table 1. Details of covariates adjusted in the multivariable analysis are also shown. It can be noted that five studies reported subgroup analysis based on different variables such as gender, age, comorbidities, and body mass index (BMI). Further, some studies also reported separate data on different short and long-sleep duration cutoffs. Such data were used in our subgroup analysis based on timings of short and long sleep.

Short sleep vs optimal sleep

Using study-specific cutoffs, we noted that short sleep was associated with a mild increase in the risk of CKD/proteinuria (OR: 1.17 95% CI 1.05, 1.31 $I^2 = 52\%$) (Fig. 2). We did not note any change in the significance of the OR on sensitivity analysis. Further, no publication bias was noted on the funnel plot (Supplementary Fig. 1) and Egger's test ($p = 0.85$).

Fig. 1 Study flowchart



Subgroup analysis for the primary outcome of short vs optimal sleep duration is presented in Table 2. The association was significant for prospective but not for retrospective studies. Similarly, significant results were seen only for Asian studies but not for Western data. On subgroup analysis based on CKD diagnosis and assessment, we noted that all subgroups demonstrated non-significant results. Importantly, the OR was borderline non-significant when CKD definition was eGFR < 60 ml/min per 1.73 m^2. Subgroup analysis of two studies including proteinuria also showed non-significant results.

Based on different cutoffs for short sleep, we noted that sleep of ≤ 7 h (OR: 1.16 95% CI 0.90, 1.48 $I^2 = 82\%$) was not associated with a significantly increased risk of CKD/proteinuria. A mild significant risk was noted in the subgroup of ≤ 6 h (OR: 1.07 95% CI 1.00, 1.14 $I^2 = 15\%$) while a significant association was noted for sleep ≤ 5 h (OR: 1.42 95% CI 1.23, 1.65 $I^2 = 0\%$) (Fig. 3). Collating data from different subgroups reported by the included studies, we noted that short sleep was not associated with CKD/proteinuria in males (OR: 1.13 95% CI 0.84, 1.53 $I^2 = 80\%$), females (OR: 1.13 95% CI 0.98, 1.30 $I^2 = 27\%$), high BMI cases (OR: 1.06 95% CI 0.98, 1.15 $I^2 = 0\%$), or elderly (≥ 60 years) individuals (OR: 1.21 95% CI 0.91, 1.62 $I^2 = 76\%$) (Fig. 4).

Long sleep vs optimal sleep

Based on study-specific cutoffs, the meta-analysis showed that long sleep duration was associated with a mild increase in the risk of CKD/proteinuria (OR: 1.14 95% CI 1.04, 1.24 $I^2 = 55\%$) (Fig. 5). There was no change in the significance of the OR on sensitivity analysis. In addition, no publication bias was noted on the funnel plot (Supplementary Fig. 2) and Egger's test ($p = 0.88$).

Details of subgroup analysis for long vs optimal sleep are shown in Table 2. Similar to the analysis of short sleep, we noted significant results only for prospective and Asian studies but not for retrospective and Western data. Studies using CKD definition of eGFR < 60 ml/min per 1.73 m^2 demonstrated significant results. However, no significant

Table 1 Details of included studies

Study	Database and duration	Included population	Design	Sample size	Age (years)	Male (%)	Classification of sleep duration	Definition of CKD	Assessment of sleep duration	Assessment of CKD	% with short sleep	% with long sleep	% with CKD	Follow-up	NOS
Jiang 2023[34]	China Health and Retirement Longitudinal Study; 2011–2018	> 45 year old without CKD	P	11,677	NR	47.3	Optimal: 7–9 h Short: < 7 h Long: > 9 h	NR	Questionnaire	Questionnaire	49.7	4.2	4.8	7 years	7
Zhang 2022[35]	UK Biobank; 2006–2010	40–69 year old without CKD	P	370,671	NR	45.4	Optimal: 7–8 h Short: < 7 h Long: > 8 h	eGFR < 60 ml/min per 1.73 m^2	Questionnaire	Medical records	24.2	7.5	1.7	10.6 years	8
Cha 2022[31]	Korean Genome and Epidemiology Study; 2001–2002	Hypertensives without CKD	P	2837	NR	49.6	Optimal: 6–9 h Short: < 6 h Long: ≥ 9 h	eGFR < 60 ml/min per 1.73 m^2	Questionnaire	Clinical measurements	16.8	9.9	25.3	9.17 ± 5.3 years	8
Nakajima 2020[32]	NAGALA study, Japan; 1994–2013	Adults without CKD and not using medications for diabetes, hypertension, dyslipidaemia, hyperuricaemia, and insomnia	R	14,474	NR	53.6	Optimal: 7–8 h Short: < 7 h Long: ≥ 8 h	Proteinuria or eGFR < 60 ml/min per 1.73 m^2	Questionnaire	Clinical measurements	61.2	7.9	15.2	6.7–7 years	8

Table 1 (continued)

Study	Database and duration	Included population	Design	Sample size	Age (years)	Male (%)	Classification of sleep duration	Definition of CKD	Assessment of sleep duration	Assessment of CKD	% with short sleep	% with long sleep	% with CKD	Follow-up	NOS
Geng 2019[33]	Singapore Chinese Health Study; 1993–1998	45–74 year old	P	63,147	NR	44	Optimal: 7 h Short: ≤ 6 h Long: ≥ 8 h	Serum creatinine level ≥ 880 μmol/L (10 mg/dl) or eGFR < 15 ml/min per 1.73 m^2 or undergoing hemodialysis or peritoneal dialysis; or has undergone kidney transplant	Questionnaire	Medical records	33	34.4	1.8	16.8 ± 5.1 years	8
Bo 2019[30]	MJ Health Management Institution, Taiwan; 1996–2014	Adults without CKD	P	194,039	38.2	50.7	Optimal: 6–8 h Short: < 6 h Long: > 8 h	eGFR < 60 ml/min per 1.73 m^2	Questionnaire	Clinical measurements	19.1	8.4	4.4	5.56 ± 4.19 years	8
McMullan 2016[29]	Nurses' Health Study, USA; 1976–2000	30–55 year old	P	4238	58	NR	Optimal: 7–8 h Short: ≤ 6 h Long: ≥ 9 h	eGRF decease of ≥ 30%	Questionnaire	Clinical measurements	27.5	4.5	12	11 years	8

Table 1 (continued)

Study	Database and duration	Included population	Design	Sample size	Age (years)	Male (%)	Classification of sleep duration	Definition of CKD	Assessment of sleep duration	Assessment of CKD	% with short sleep	% with long sleep	% with CKD	Follow-up	NOS
Sasaki 2014[28]	Sapporo, Hokkaido Prefecture, Japan; 2003–2008	35–59 year old without CKD and not undergoing treatment for hypertension, diabetes mellitus, dyslipidemia, or renal disease and no history of coronary artery disease or stroke	P	3600	47	77.7	Optimal: 6–7 h Short: ≤ 5 h Long: ≥ 8 h	eGFR < 60 ml/min per 1.73 m^2	Questionnaire	Clinical measurements	8.2	19	5	4.4 years	8
Yamamoto 2012[27]	Osaka University, Japan; 2006–2010	20–65 year old without CKD	R	6834	34	50.4	Optimal: ≥ 7 h Short: ≤ 6 h	Proteinuria, defined as urinary protein (≥ 1 +) by dipstick test	Questionnaire	Clinical measurements	72.6	-	8	NR	7

CKD Chronic kidney disease, *P* Prospective, *R* Retrospective, *eGFR* Estimated glomerular filtration rate, *h* Hours, *NR* Not reported, *NOS*, Newcastle ottawa scale

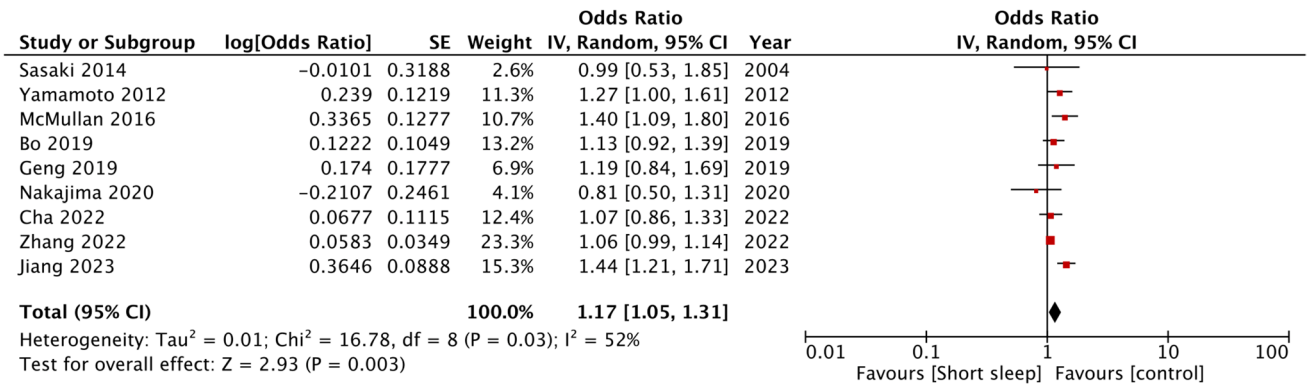


Fig. 2 Meta-analysis of the association between short sleep and risk of CKD/proteinuria

Table 2 Subgroup analysis

Outcome	Variable	Groups	Studies	Odds ratio [95% confidence interval]	I^2
Short sleep duration	Study design	Prospective	7	1.18 [1.05, 1.33]	57
		Retrospective	2	1.07 [0.69, 1.64]	63
	Location	Asian	7	1.19 [1.05, 1.34]	31
		Western	2	1.19 [0.91, 1.55]	77
	CKD definition	eGFR < 60 ml/min per 1.73 m^2	4	1.07 [1.00, 1.13]	0
		Including proteinuria	2	1.07 [0.69, 1.64]	63
	CKD assessment	Clinical measurements	6	1.16 [1.03, 1.31]	12
		Questionnaire/records	3	1.21 [0.96, 1.52]	81
Long sleep duration	Study design	Prospective	7	1.18 [1.09, 1.27]	38
		Retrospective	1	0.97 [0.83, 1.13]	-
	Location	Asian	6	1.11 [1.01, 1.21]	40
		Western	2	1.18 [0.89, 1.57]	37
	CKD definition	eGFR < 60 ml/min per 1.73 m^2	4	1.15 [1.03, 1.28]	60
		Including proteinuria	1	0.97 [0.83, 1.13]	-
	CKD assessment	Clinical measurements	5	1.06 [0.97, 1.16]	17
		Questionnaire/records	3	1.26 [1.18, 1.35]	0

eGFR Estimated glomerular filtration rate, *CKD* Chronic kidney disease, *ESRD* End-stage renal disease

association was noted for proteinuria and CKD. Subgroup analysis based on CKD assessment showed a positive association for studies using medical records/questionnaires but non-significant results for those using clinical measurements.

Two subgroups were examined based on the duration of long sleep, namely ≥ 8 h and ≥ 9 h. Comparing with study-specific optimal sleep, individuals with ≥ 8 h of sleep had an increased risk of CKD/proteinuria (OR: 1.14 95% CI 1.04, 1.25 $I^2 = 62\%$). However, the OR was not significant for individuals with ≥ 9 h of sleep (OR: 1.14 95% CI 0.92, 1.41 $I^2 = 41\%$) (Fig. 6). Combining data of subgroups reported by individual studies, we noted that long sleep duration was not associated with an increased risk of CKD/proteinuria in males (OR: 1.15 95% CI: 0.98, 1.34 $I^2 = 62\%$), females (OR: 1.05 95% CI 0.97, 1.14 $I^2 = 0\%$), high BMI cases (OR: 1.10 95% CI 0.99, 1.22 $I^2 = 0\%$), or elderly (≥ 60 years) individuals (OR: 1.13 95% CI 0.98, 1.30 $I^2 = 0\%$) (Fig. 7).

Discussion

We have presented the first meta-analysis of cohort studies examining the association between sleep duration and long-term risk of CKD/proteinuria. A detailed literature search across multiple databases was undertaken to retrieve nine studies published in the past 2 decades. Combining data from 671,517 participants, we found that there was a mild increased risk of CKD/proteinuria with both short and long sleep duration. However, the results were not consistent across subgroups of males, females, high BMI, and elderly (≥ 60 years). Further, no consistent trend was noted with different cutoffs of short and long sleep.

Sleep is crucial in maintaining physiological function and homeostasis as it provides a window for the repair and regeneration of tissues. An average adult spends about

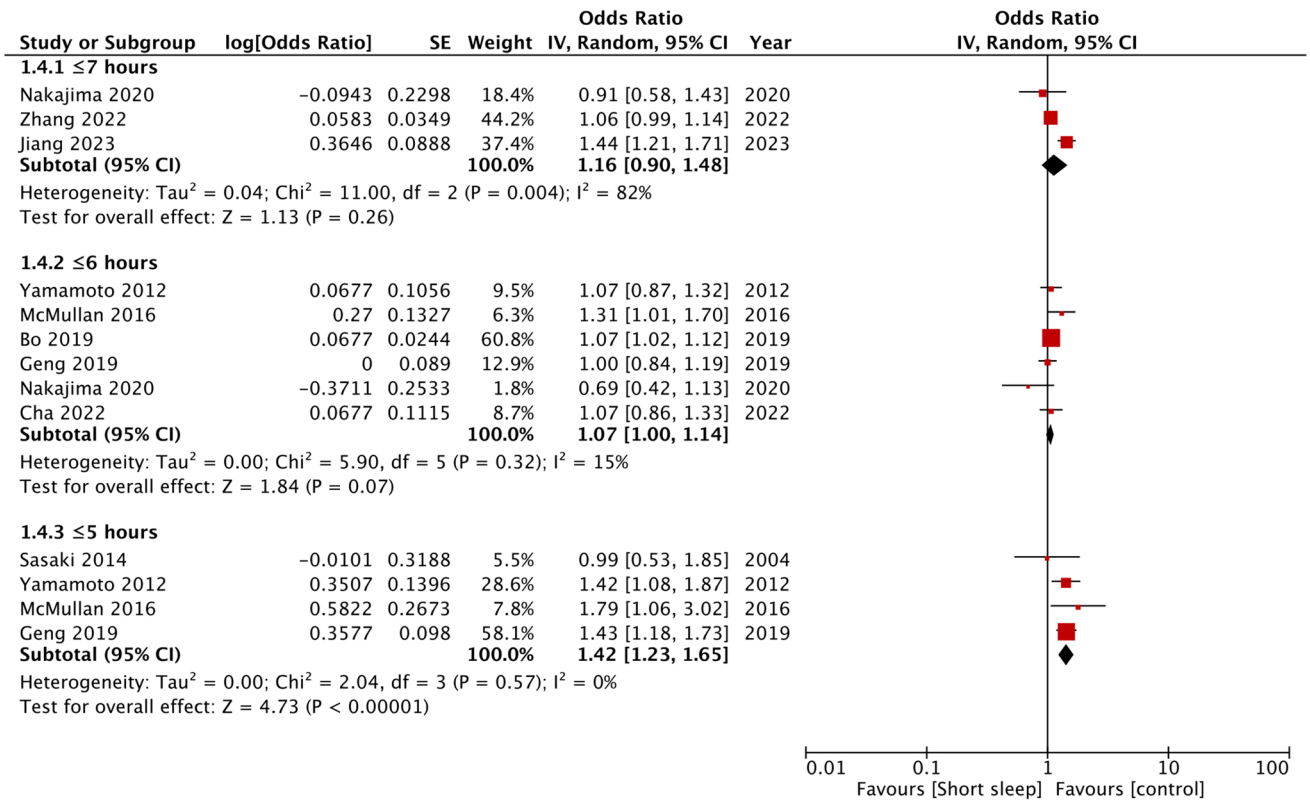


Fig. 3 Meta-analysis of the association between short sleep and risk of CKD/proteinuria with different cutoffs of short sleep

one-third of his or her lifetime sleeping [2, 3]. However, long working hours, prolonged travel, and stressful environments have become a norm in modern society which have significantly reduced sleep quality and duration [36]. Therefore, sleep duration has been recognized as an important lifestyle risk factor, particularly concerning short sleep duration [1, 6–9]. A large meta-analysis of 153 cohort studies has shown that short sleep increases the risk of all-cause mortality, diabetes, hypertension, cardiovascular disease, coronary heart disease, and obesity by 12%, 37%, 17%, 16%, 26%, and 38%, respectively [1]. Similarly, long sleep (≥ 9 h) has been highly prevalent (> 25%) in several developed countries [37] and it is also a recognized risk factor for systemic diseases increasing the probability of mortality, diabetes, cardiovascular disease, stroke, coronary heart disease, and obesity by 39%, 26%, 25%, 46%, 24% and 8%, respectively [10]. An identical relationship between short and long sleep duration and CKD/proteinuria has been suggested previously but with conflicting results. The first meta-analysis of Cheungpasitporn et al. [21] in 2017 combined data from two cohort and six cross-sectional studies to find a non-significant association between short sleep and CKD (RR: 1.54 95% CI 0.80, 2.95) but there was a significant relationship between short sleep and proteinuria (RR: 1.47 95% CI 1.26–1.72). The second meta-analysis by Hao et al. [22] published in 2020 included three cohort and eight cross-sectional studies and reported that both short and long sleep were predictors of CKD and proteinuria. In addition to the low number of cohort studies, there were several other limitations to this review such as inclusion of studies with unadjusted data and studies on specific populations of type-2 diabetes and hypertension, both of which are independent risk factors of CKD [17, 38]. Further, a recent meta-analysis by Koh et al. [39] have also examined the same clinical question and reported increased risk of incident CKD with both short and long sleep duration. While the authors claimed to have included 14 cohort studies in each analysis, several cross-sectional studies were wrongly identified as cohort studies and used in the meta-analysis of incident CKD.

The present meta-analysis overcomes these limitations and presents high-quality evidence based on only cohort studies reporting adjusted data. A large number of confounders were adjusted by the included studies while exploring the relationship between sleep duration and CKD/proteinuria. Given that CKD has a slow progression, long longitudinal follow-up is necessary to establish a link between sleep and CKD/proteinuria. All included studies in this review reported a long follow-up ranging from 4 to 16 years. Overall, the meta-analysis showed that short and long sleep was

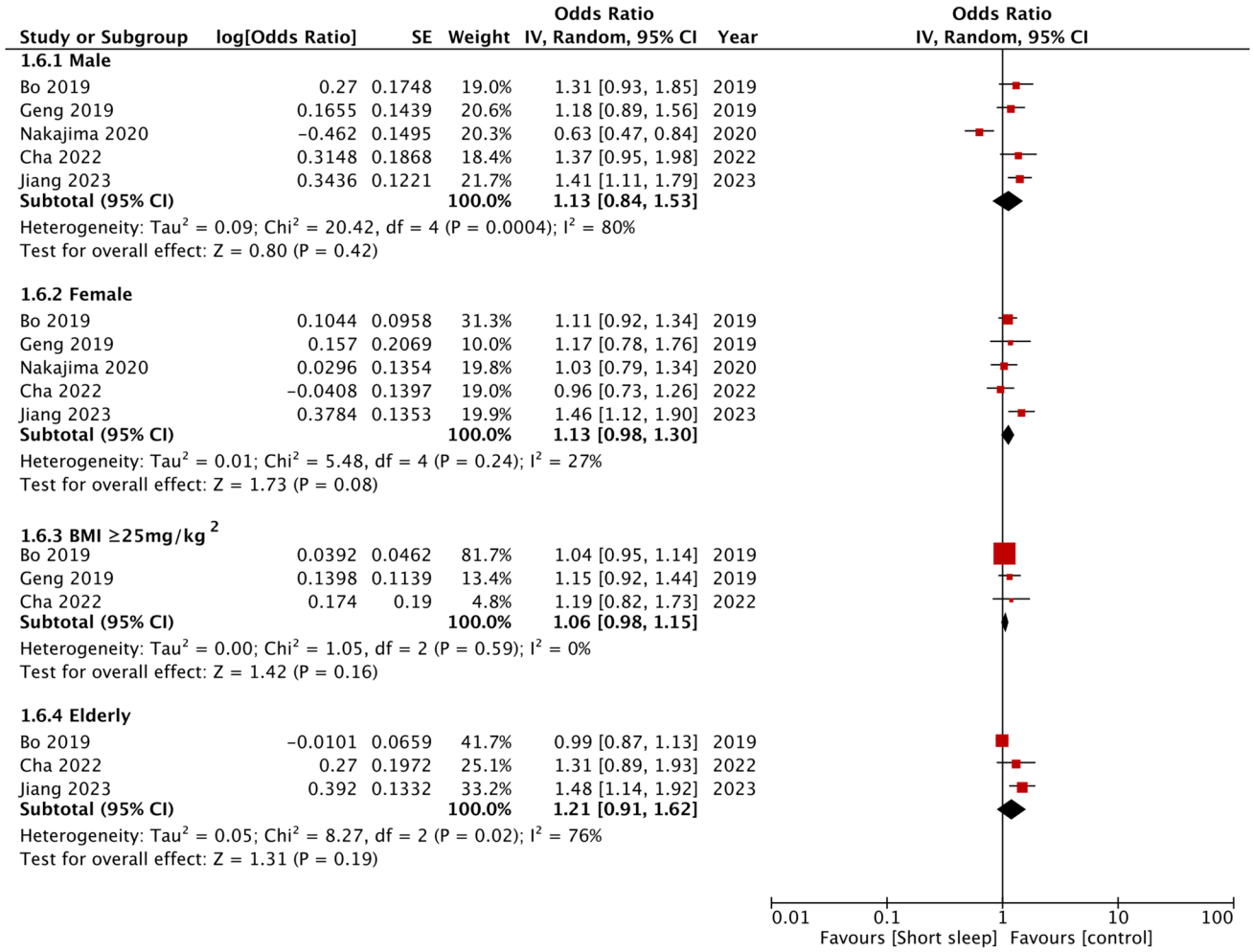


Fig. 4 Meta-analysis of the association between short sleep and risk of CKD/proteinuria in specific cohorts of males, females, high BMI and elderly

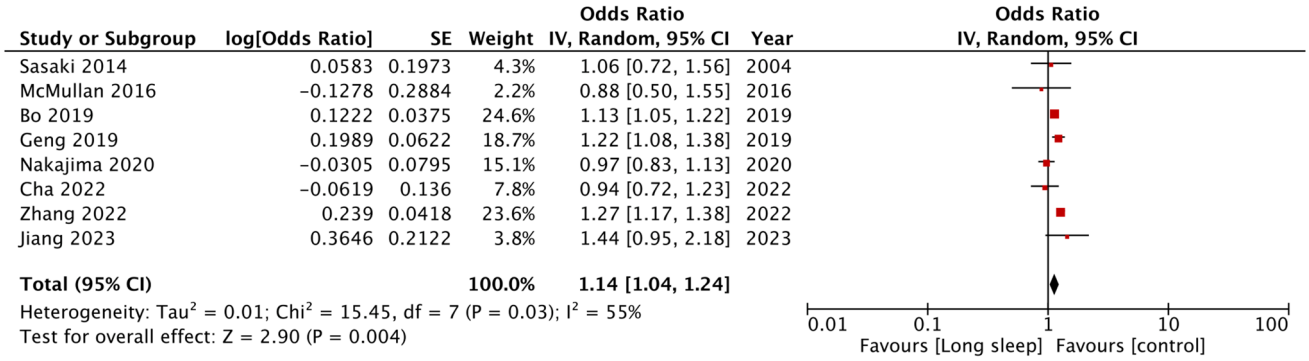


Fig. 5 Meta-analysis of the association between long sleep and risk of CKD/proteinuria

associated with a significant but only modest increase in the risk of CKD/proteinuria (17% and 14%, respectively). Nevertheless, the association was found to be robust in sensitivity analysis. On the other hand, the high heterogeneity is a cause of concern and attempts to explore the cause of heterogeneity led to mixed results. We noted that prospective studies favored the association while lower-quality retrospective studies did not. Similarly, the association was significant

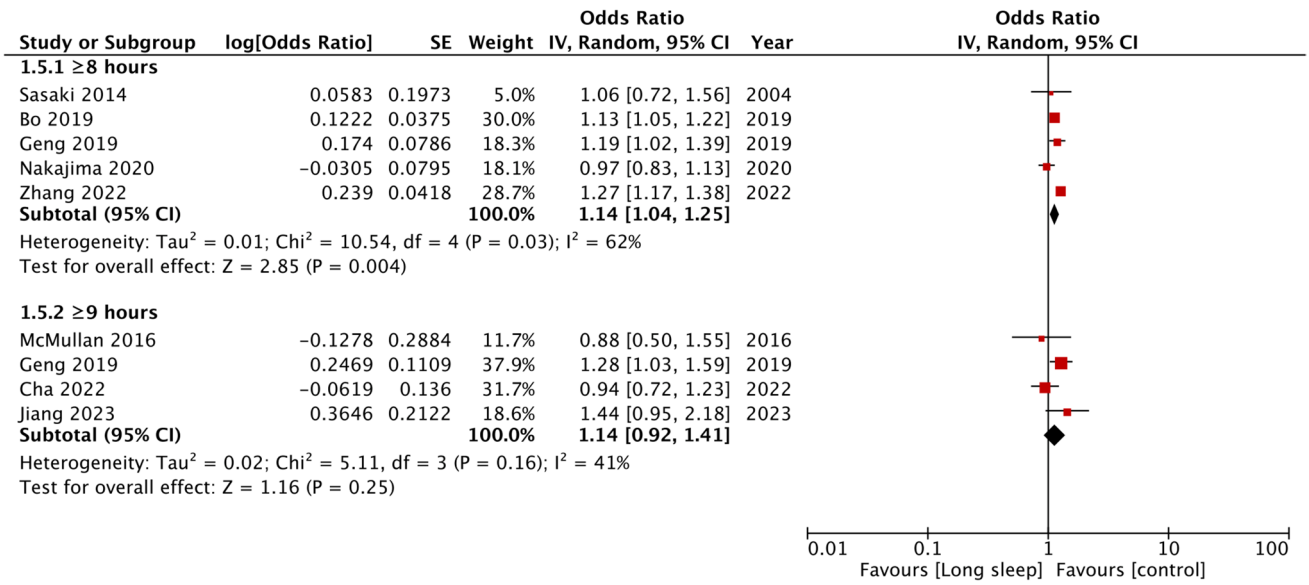


Fig. 6 Meta-analysis of the association between long sleep and risk of CKD/proteinuria with different cutoffs of short sleep

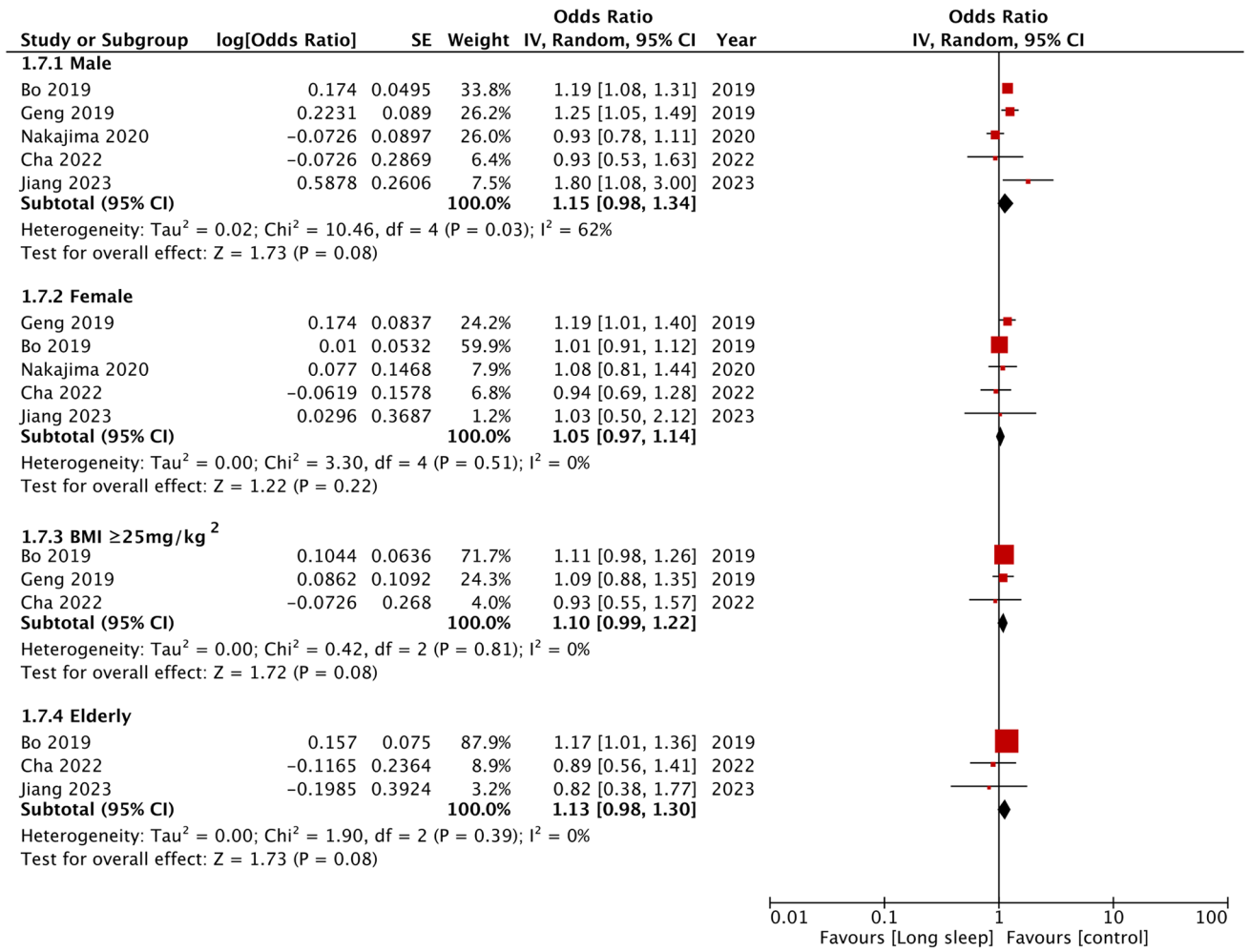


Fig. 7 Meta-analysis of the association between long sleep and risk of CKD/proteinuria in specific cohorts of males, females, high BMI and elderly

only for Asian populations and not for Western countries. One reason for this discrepancy could be the predominance of Asian data in the systematic review. Second, there was homogeneity in the method of assessment of sleep duration but not for CKD. Using the standard definition of CKD of eGFR < 60 ml/min per 1.73 m^2 [15], we found a non-significant association between short sleep and CKD but a statistically significant 15% increase in the risk of CKD with long sleep. No significant association was noted between short and long sleep when the CKD definition included proteinuria. The later results were generated by too few studies and need to be supplemented by further research.

Gender, BMI, and age are important confounders in the incidence of CKD. Gender has a complex relationship with CKD with females having a higher prevalence of CKD and males being more commonly diagnosed with ESKD. Males carry a higher burden of CKD risk factors such as hypertension and obesity while females have limited access to primary renal care [40]. Further, obese and elderly individuals are at a particularly higher risk of CKD [41, 42]. Analyzing specific subgroups based on gender, BMI, and age, we noted that short or long sleep was not associated with an increased risk of CKD/proteinuria in males, females, high BMI, and elderly groups. A consistent positive but non-significant trend was noted for all these subgroups albeit with the lower end of 95% CI very near to 1. This could be due to the limited sample size of these groups and the smaller number of studies in the meta-analysis which failed to reach statistical significance.

A major variation noted in the included studies was in the definition of short, long, and optimal sleep which is a major source of bias. While guidelines suggest that sleep of 7–9 h is optimal [2, 3], there is no universally adopted definition of short or long sleep [43]. Previous reviews and meta-analysis studies have also faced similar problems in defining the optimal cutoff for both short and long sleep [1, 10, 22]. To overcome this issue, we conducted separate analyses based on specific cutoffs of short and long sleep. In the case of short sleep, we noted an inverse relationship between the duration of short sleep and the risk of CKD/proteinuria. No significant association was noted between ≤ 7 h of sleep and the risk of CKD/proteinuria while a 6% borderline significant increase in the risk of CKD/proteinuria was noted with ≤ 6 h of sleep. Lastly, ≤ 5 h of sleep showed the strongest association with the highest risk of CKD/proteinuria (42%). However, a similar association was not noted with long sleep duration as ≥ 8 h of sleep demonstrated a significant association with CKD/proteinuria but the subgroup of ≥ 9 h of sleep could not achieve significant results. The ambiguous results of the latter could be due to the small number of studies in the subgroup analysis and variations in the definitions of optimal sleep duration, CKD, or proteinuria. However, given that all ORs were still > 1, we believe that a positive relationship could still exist between long sleep duration and CKD/proteinuria but a specific cutoff is yet to be clearly defined. In case of short sleep, the results suggest that at least 7 h of sleep is needed to offset the risk of CKD/proteinuria, nevertheless, these results are obtained from a small number of cohort studies and need to be further validated.

Several potential mechanisms can justify the relationship between sleep duration and CKD. Sleep deprivation causes sympathetic hyperactivity which increases the risk of CKD by its effect on blood pressure and vascular tone [44, 45]. Sleep deprivation induces excess nocturnal natriuresis and osmotic diuresis with altered activity of the renin–angiotensin–aldosterone system which is critical in the development of CKD [46, 47]. Short sleep results in a higher degree of systemic inflammation with an increase in inflammatory and cardiovascular biomarkers such as C-reactive protein, lipoproteins, triglycerides, glucose, and insulin which contribute to a higher risk of cardiovascular disorders and in turn CKD [48]. On the other hand, long sleep duration has been associated with physical inactivity, a dysfunctional immune system, and chronic inflammation which could also contribute to increased risk of CKD [49, 50]. Individuals with longer sleep often experience sleep fragmentation and poor sleep quality [51]. Sleep fragmentation causes changes in the immune system and could lead to poor overall health outcomes [52].

We need to acknowledge some limitations of this review. First, sleep assessment was self-reported in all studies and not measured objectively using actigraphy or polysomnography. Self-reported sleep can be biased and inconsistent and potentially introduce bias in the results. Second, we focused only on sleep duration and not on sleep quality as it is one of the most easily recognized and accurately reported sleep components [10]. Nevertheless, sleep components such as onset latency, sleep efficiency, number of arousals, and daytime sleepiness are important variables that could impact the association between sleep duration and CKD/proteinuria. Third, while we used only adjusted data from the studies, the type of confounders was completely dependent on the individual studies. Inconsistencies among studies meant that several known and unknown confounders could have been during the analysis of the association. Fourth, the variability in the definition of CKD is a major limitation. Some studies measured a decrease in eGFR, some assessed only the risk of ESKD, while some measured only proteinuria. We circumvented this issue using a subgroup analysis but with reduced statistical power of the analysis. Fifthly, there were other methodological variations among studies such as the included population, definition of sleep durations, assessment of CKD, and follow-up which could have skewed the results. Not all studies excluded individuals based on baseline CKD, comorbidities, risk factors of CKD, and use of

medications for insomnia. One study [31] was on hypertensives alone while others were on the general population. While the results did not change on the exclusion of this study [31] on sensitivity analysis, the inclusion of heterogenous populations by the included studies is a serious limitation. Sixth, the included studies did not use a common effect size and IRR, hazard ratio, and OR were combined in a single meta-analysis. It is pertinent to note that the hazard ratio is a time-dependent measure that takes loss to follow-up and time at risk into account, but OR and risk ratios assume a single time point. As follow-up varied across studies, there is no way to know if the estimates were exactly comparable. Such inconsistencies in effect size have also been noted by other meta-analyses [1, 9, 22] on sleep duration and different ratios were combined in these reviews too. Lastly, most data were from Asian studies and, therefore, the findings cannot be generalized.

Conclusions

The meta-analysis of good-quality cohort studies reveals that short and long sleep durations are associated with a significant increase in the risk of CKD/proteinuria in the adult population. No specific cutoff of short or long sleep was found to be predominantly associated with CKD/proteinuria. Future studies should explore if sleep hygiene interventions can modify the risk of CKD. In addition, there is a need for studies using similar cutoff definitions and examining non-Asian populations to further strengthen the current results.

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Data availability Data are provided within the manuscript or supplementary information files.

Declarations

Conflict of interests The authors declare no competing interests.

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